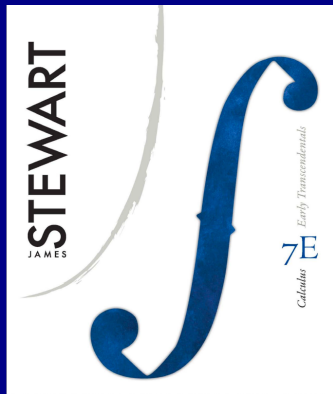




MATH1560J RC1

1. Reciprocal Trigonometric Functions

Core Definitions

The three reciprocal trigonometric functions are defined based on sin, cos, and tan.

- **Secant (sec):** The reciprocal of cos.

$$\sec(x) = \frac{1}{\cos(x)}$$

- **Cosecant (csc):** The reciprocal of sin.

$$\csc(x) = \frac{1}{\sin(x)}$$

Core Definitions

- **Cotangent (cot):** The reciprocal of tangent.

$$\cot(x) = \frac{1}{\tan(x)} = \frac{\cos(x)}{\sin(x)}$$

Memorization Tips

In each reciprocal pair, if one function starts with an 's', then its reciprocal partner starts with a 'c'.

- $\sin \leftrightarrow \csc$
- $\cos \leftrightarrow \sec$

The Fundamental Identity

This is the most important trigonometric identity, derived from the unit circle:

$$\sin^2(x) + \cos^2(x) = 1$$

Derived Identities

By dividing the fundamental identity by $\cos^2(x)$ and $\sin^2(x)$, we get two other key identities.

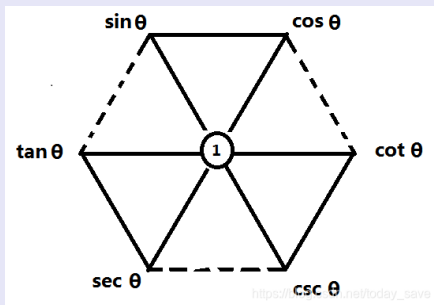
- ① Dividing by $\cos^2(x)$ yields:

$$\tan^2(x) + 1 = \sec^2(x)$$

- ② Dividing by $\sin^2(x)$ yields:

$$1 + \cot^2(x) = \csc^2(x)$$

This hexagon is a powerful tool to remember many trigonometric identities!



Basic Rules

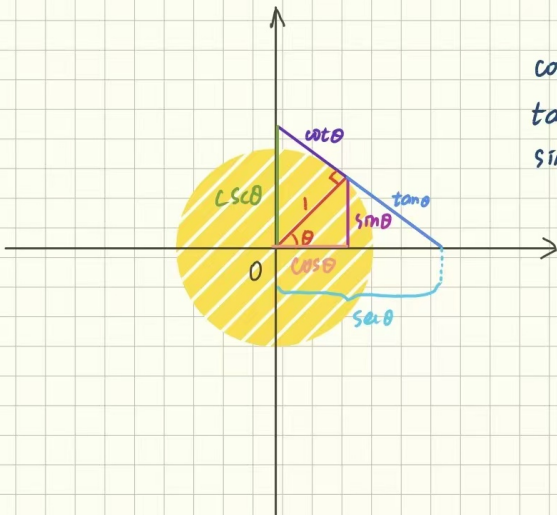
- Functions on opposite ends of a diagonal are **reciprocals**.
 - $\sin(x) \longleftrightarrow \csc(x)$
 - $\cos(x) \longleftrightarrow \sec(x)$
- Any function is the **product** of its two neighbors.
 - $\sin(x) = \tan(x) \cdot \cos(x)$



Focus on the three shaded triangles. The sum of the squares of the top two vertices equals the square of the bottom vertex.

$$(\text{Top Left})^2 + (\text{Top Right})^2 = (\text{Bottom})^2$$

- **Top Triangle:**
 $\sin^2(x) + \cos^2(x) = 1^2$
- **Bottom Left Triangle:**
 $1^2 + \cot^2(x) = \csc^2(x)$
- **Bottom Right Triangle:**
 $\tan^2(x) + 1^2 = \sec^2(x)$



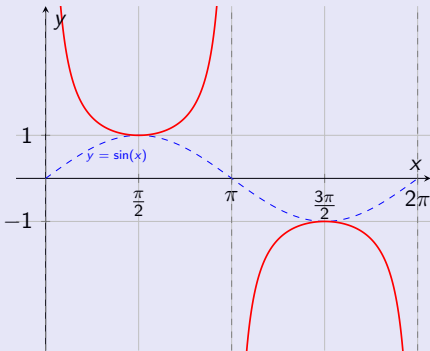
$$\cot^2 \theta + 1 = \csc^2 \theta$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

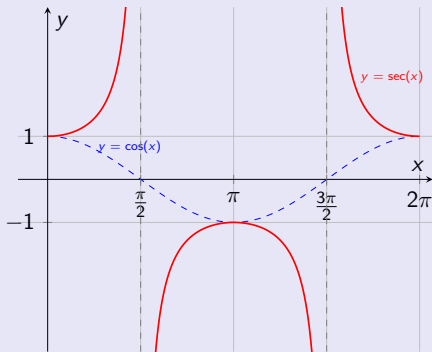
Graphing $y = \csc(x)$ from $y = \sin(x)$ **Method:**

- 1 Sketch $y = \sin(x)$.
- 2 Where $\sin(x) = 0$, draw vertical asymptotes for $\csc(x)$.
- 3 Where $\sin(x) = \pm 1$, the graphs touch.
- 4 Sketch the 'U' shapes between asymptotes.



Method:

- 1 Sketch $y = \cos(x)$.
- 2 Where $\cos(x) = 0$, draw vertical asymptotes for $\sec(x)$.
- 3 Where $\cos(x) = \pm 1$, the graphs touch.
- 4 Sketch the 'U' shapes between asymptotes.



Method:

- 1 Sketch $y = \tan(x)$ (or consider $\sin(x)/\cos(x)$).
- 2 Where $\sin(x) = 0$, draw vertical asymptotes for $\cot(x)$.
- 3 Note that $\cot(x) = 0$ where $\cos(x) = 0$.
- 4 Sketch the 'S' shapes between asymptotes (period π).

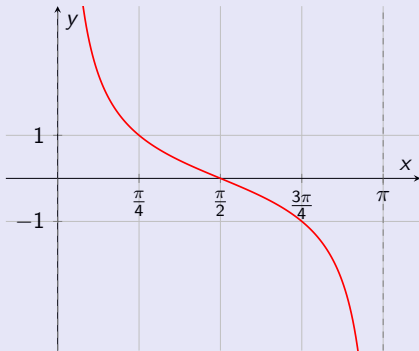
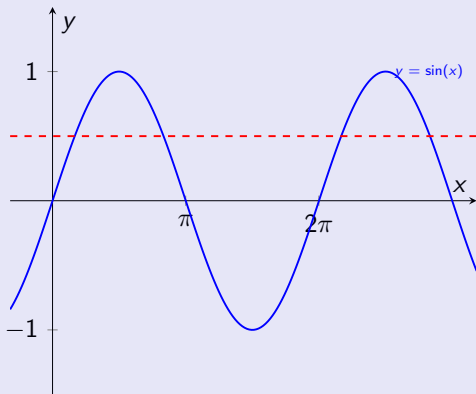


Table: Properties of Reciprocal Trigonometric Functions

Function	Domain ($x \neq \dots$)	Range	Period
$y = \csc(x)$	$n\pi$	$(-\infty, -1] \cup [1, \infty)$	2π
$y = \sec(x)$	$\frac{\pi}{2} + n\pi$	$(-\infty, -1] \cup [1, \infty)$	2π
$y = \cot(x)$	$n\pi$	$(-\infty, \infty)$	π

2. Inverse Trigonometric Functions



Problem

A function must be **one-to-one** to have an inverse. $y = \sin(x)$ fails the Horizontal Line Test.

Solution

Restrict the domain to a piece where the function is one-to-one.

- For $\sin(x)$, we use $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

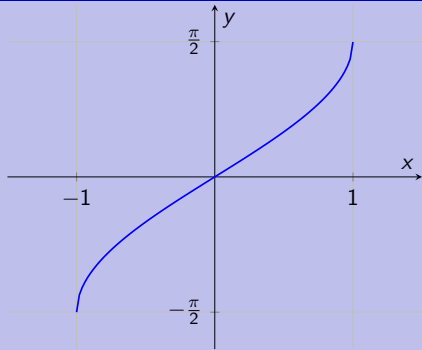
Inverse Sine & Cosine: Definition and Graph

$$y = \arcsin(x)$$

Definition

We restrict $y = \sin(x)$ to $[-\frac{\pi}{2}, \frac{\pi}{2}]$.
 $\arcsin(x)$ is the angle y in this range whose $\sin$ is x .

Graph

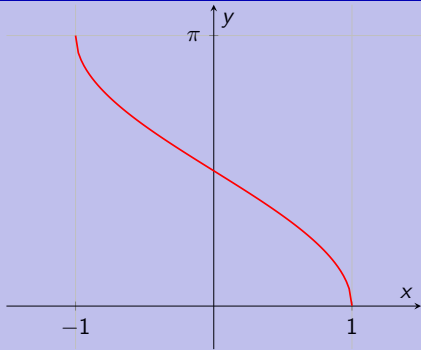


$$y = \arccos(x)$$

Definition

We restrict $y = \cos(x)$ to $[0, \pi]$.
 $\arccos(x)$ is the angle y in this range whose $\cos$ is x .

Graph



$$y = \arcsin(x)$$

$$y = \arccos(x)$$

Properties

- **Domain:** $[-1, 1]$
- **Range:** $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- **Derivative:** $\frac{d}{dx} = \frac{1}{\sqrt{1-x^2}}$

Properties

- **Domain:** $[-1, 1]$
- **Range:** $[0, \pi]$
- **Derivative:** $\frac{d}{dx} = -\frac{1}{\sqrt{1-x^2}}$

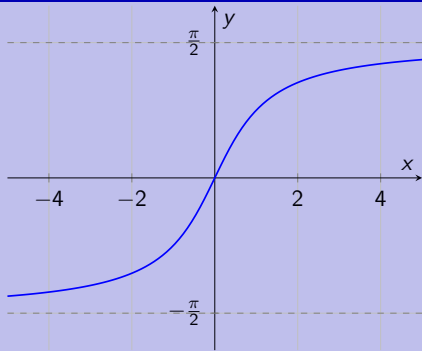
Inverse Tangent & Cotangent: Definition and Graph

$$y = \arctan(x)$$

Definition

We restrict $y = \tan(x)$ to $(-\frac{\pi}{2}, \frac{\pi}{2})$.
 $\arctan(x)$ is the angle y in this range whose tangent is x .

Graph

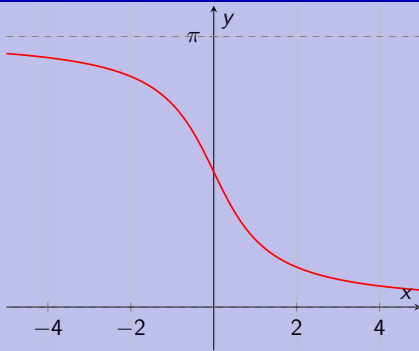


$$y = \operatorname{arccot}(x)$$

Definition

We restrict $y = \cot(x)$ to $(0, \pi)$.
 $\operatorname{arccot}(x)$ is the angle y in this range whose cotangent is x .

Graph



$$y = \arctan(x)$$

$$y = \operatorname{arccot}(x)$$

Properties

- **Domain:** $(-\infty, \infty)$
- **Range:** $(-\frac{\pi}{2}, \frac{\pi}{2})$
- **Derivative:** $\frac{d}{dx} = \frac{1}{1+x^2}$

Properties

- **Domain:** $(-\infty, \infty)$
- **Range:** $(0, \pi)$
- **Derivative:** $\frac{d}{dx} = -\frac{1}{1+x^2}$

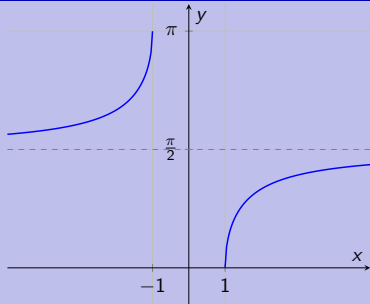
Inverse Secant & Cosecant: Definition and Graph

$$y = \operatorname{arcsec}(x)$$

Definition

We restrict $y = \sec(x)$ to $[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$. $\operatorname{arcsec}(x)$ is the angle y in this range whose secant is x .

Graph

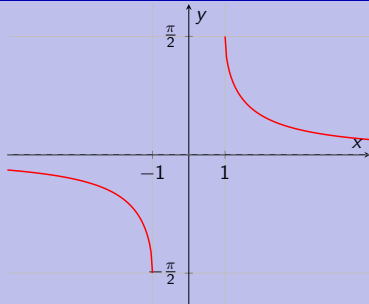


$$y = \operatorname{arccsc}(x)$$

Definition

We restrict $y = \csc(x)$ to $[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$. $\operatorname{arccsc}(x)$ is the angle y in this range whose cosecant is x .

Graph



$$y = \operatorname{arcsec}(x)$$

$$y = \operatorname{arccsc}(x)$$

Properties

- **Domain:** $|x| \geq 1$
- **Range:** $[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$
- **Derivative:** $\frac{d}{dx} = \frac{1}{|x|\sqrt{x^2-1}}$

Properties

- **Domain:** $|x| \geq 1$
- **Range:** $[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$
- **Derivative:** $\frac{d}{dx} = -\frac{1}{|x|\sqrt{x^2-1}}$

The output of an inverse trigonometric function is an **angle**. We must give the answer within the standard, restricted range (the principal value).

Function	Domain of x	Range of y (Angle)
$y = \arcsin(x)$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$y = \arccos(x)$	$[-1, 1]$	$[0, \pi]$
$y = \arctan(x)$	$(-\infty, \infty)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$y = \operatorname{arccot}(x)$	$(-\infty, \infty)$	$(0, \pi)$
$y = \operatorname{arcsec}(x)$	$ x \geq 1$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$
$y = \operatorname{arccsc}(x)$	$ x \geq 1$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$

Warning!

$\sin^{-1}(x)$ means inverse sine, NOT $\frac{1}{\sin(x)}$. Different textbooks may use different range conventions for arcsec and arccsc .

3. Hyperbolic Functions

Hyperbolic functions are analogues of trigonometric functions, but are defined using the exponential function e^x .

Hyperbolic Cosine (cosh)

The **even** part of e^x . Pronounced "gosh".

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

Hyperbolic Sine (sinh)

The **odd** part of e^x . Pronounced "cinch".

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

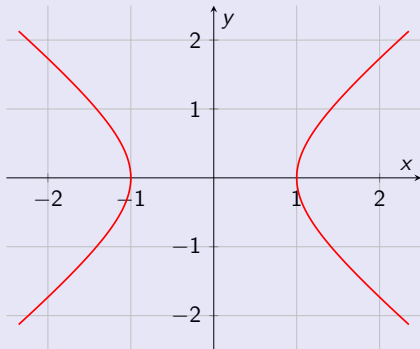
The Fundamental Identity

$$\cosh^2(x) - \sinh^2(x) = 1$$

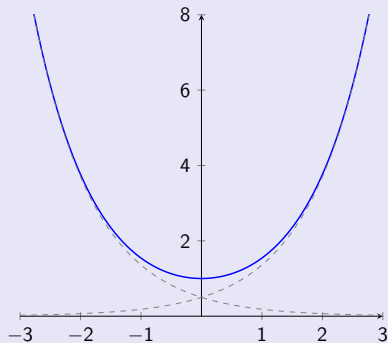
Geometric Meaning

- A point $(\cos(t), \sin(t))$ lies on the unit **circle**: $x^2 + y^2 = 1$.
- A point $(\cosh(t), \sinh(t))$ lies on the unit **hyperbola**: $x^2 - y^2 = 1$.

Unit Hyperbola $x^2 - y^2 = 1$



$$y = \cosh(x) \text{ (Even)}$$



$$y = \sinh(x) \text{ (Odd)}$$

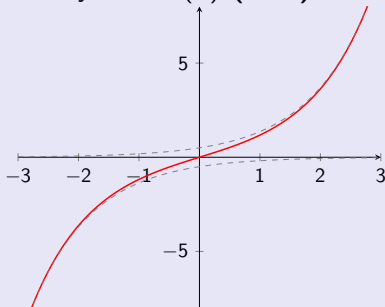




Figure: The shape of a hanging chain or cable is not a parabola, but a **catenary**.

The Catenary Equation

The curve is described by the hyperbolic cosine function:

$$y = a \cosh \left(\frac{x}{a} \right)$$

This shape appears in physics and engineering, from power lines to architecture.

Today We Covered:

- **Reciprocal Functions (sec, csc, cot):** Definitions, graphs, and key properties.
- **Inverse Functions (all six):** The need for restricted domains and how to evaluate expressions.
- **Hyperbolic Functions (sinh, cosh):** Definitions from e^x , the core identity, and the catenary application.

Any Questions?

